

Lab 8



- Implement the Dijkstra's shortest path algorithm.
(Find the shortest paths between a single source node and all the other nodes)

```
#include <stdio.h>
#include <stdlib.h>

int INF = 9999999;

void ShortestPath(int graph[5][5], int length[5], int numNode) {

}

int main() {
    int graph[5][5] = {{INF, 10, 5, INF, INF},
                       {INF, INF, INF, 1, INF},
                       {INF, 3, INF, 9, 2},
                       {INF, INF, INF, INF, 4},
                       {7, INF, INF, 6, INF}};
    int length[5];
    ShortestPath(graph, length, 5);
    int i = 0;
    for (i = 1; i < 5; i++) {
        printf("The shortest path between V0 and V%d is %d\n", i, length[i]);
    }
    return 0;
}
```

```

void ShortestPath_DIJ( MGraph G, int v0, PathMatrix &P, ShortPathTable &D ) {
    // 用 Dijkstra 算法求有向网 G 的 v0 顶点到其余顶点 v 的最短路径 P[v] 及其带权长度 D[v]。
    // 若 P[v][w] 为 TRUE, 则 w 是从 v0 到 v 当前求得最短路径上的顶点。
    // final[v] 为 TRUE 当且仅当 v ∈ S, 即已经求得从 v0 到 v 的最短路径。
    for (v = 0; v < G.vexnum; ++v) {
        final[v] = FALSE; D[v] = G.arcs[v0][v];
        for (w = 0; w < G.vexnum; ++w) P[v][w] = FALSE; // 设空路径
        if (D[v] < INFINITY) {P[v][v0] = TRUE; P[v][v] = TRUE; }
    } // for
    D[v0] = 0; final[v0] = TRUE; // 初始化, v0 顶点属于 S 集
    // 开始主循环, 每次求得 v0 到某个 v 顶点的最短路径, 并加 v 到 S 集
    for (i = 1; i < G.vexnum; ++i) { // 其余 G.vexnum - 1 个顶点
        min = INFINITY; // 当前所知离 v0 顶点的最近距离
        for (w = 0; w < G.vexnum; ++w)
            if (!final[w]) // w 顶点在 V - S 中
                if (D[w] < min) {v = w; min = D[w]; } // w 顶点离 v0 顶点更近
        final[v] = TRUE; // 离 v0 顶点最近的 v 加入 S 集
        for (w = 0; w < G.vexnum; ++w) // 更新当前最短路径及距离
            if (!final[w] && (min + G.arcs[v][w] < D[w])) { // 修改 D[w] 和 P[w], w ∈ V - S
                D[w] = min + G.arcs[v][w];
                P[w] = P[v]; P[w][w] = TRUE; // P[w] = P[v] + [w]
            } // if
    } // for
} // ShortestPath_DIJ

```

Lab 8 Submission

Due at 23:59, May 24.

Note: Submission after 23:59, May 24 will NOT be graded.

Zip file name: StudentID_LastName_FirstName_Lab#
(Please use Zip to compress your submission)

Replace # with the lab number.

Assume my ID is 12345678, the zip file name for Lab 8 is:

12345678_Liang_Pengpeng_Lab8

Note: If you do not follow the rule of the name of the Zip file, I might not be able to find your submission.

Send to: Golam Rabbi
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